

# INDIAN ECONOMY

LEVEL  
II



## LECTURES

- A BRIEF ON FISHERIES SECTOR
- A BRIEF ON APICULTURE
- A BRIEF ON PLANTATION CROPS

MODULE - 67





## A BRIEF ON FISHERIES SECTOR

- India is the second largest fish producing country in the world accounting for 7.56% of global production.
- It contributes about 1.24% to the country's GVA, and over 7.28% to the agricultural GVA.
- Fisheries demonstrated an outstanding double digit average annual growth rate of around 11% since 2014-15, with record fish production of 145 lakh tonnes in FY 2020-21.
- The Government of India in 2018-19 extended the facility of KCC to fisheries also, in addition to animal husbandry farmers. The credit limit for new KCC holders for fisheries is Rs. 2 lakh.

- Government also launched a flagship scheme of Rs. 20,050 crores, called Pradhan Mantri Matsya Sampada Yojana (PMMSY) in May 2020.
- Under PMMSY, the interventions include
  - ✓ Enhancing fish production and productivity.
  - ✓ Modernizing and strengthening the value chain.
  - ✓ Creating fisheries and post harvest infrastructure.
  - ✓ Developing robust fisheries management and regulatory frameworks.

**Fisheries provide livelihood to about 16 million fishermen and fish farmers in the country.** 



# SIGNIFICANT GROWTH IN INLAND FISHERIES ... WHEREAS MARINE FISHERIES ARE ALMOST STAGNANT!



***Andhra Pradesh leads in overall fisheries production and around 30% comes from Andhra Pradesh. The productivity of India's inland fisheries is just 3 tonnes per hectare!***

- The Government created separate Ministry for Fisheries, Animal Husbandry and Dairying. India is the second largest producer of fisheries after China.
- Andhra Pradesh leads in inland fisheries, much ahead of other States.
- When one looks at marine sector, Gujarat leads and Andhra Pradesh comes second. We can conclude that Gujarat and Andhra Pradesh account for around 40% of marine fisheries production.
- The production of inland fisheries is more than double in comparison to marine fisheries.



# TESTING OF HONEY SAMPLES BY NMR SPECTROSCOPY WAS IN THE NEWS!



*Efforts are being made to boost honey output through the “Sweet Kranti” Mission introduced in May 2017.*

- Study of samples of select Indian Brands of Honey in 2020 prima facie suggested that they are liberally adulterated with cheap sugar syrup, imported from China.
- They passed Indian tests. These samples seemed fine after an analysis by a premier National Dairy Development Board (NDDB) facility.
- However, when 13 honey samples were tested by using the German technique, Nuclear Magnetic Resonance (NMR) Spectroscopy, majority of them failed.
- The controversy raises quality concerns over a product that is widely used as an immunity booster and prophylactic, more so in the context of COVID-19.



## APICULTURE BENEFITS THE NATION IN MANY WAYS!

### 2019 REPORT BY THE ECONOMIC ADVISORY COUNCIL TO THE PM!

Around four lakh people are involved in beekeeping. They produce 110 to 115 thousand tonnes of honey every year from 34 lakh bee colonies.



### 73% OF THE CULTIVATED CROPS ARE POLLINATED BY BEES!

The report by the EAC points out that approximately 73% of the world's cultivated crops are pollinated by bees, 19% by flies, 6.5% by bats, 5% by wasps and 5% by beetles.





## HOW DOES THE AGRICULTURE SECTOR BENEFIT FROM APICULTURE?

- As per the EAC report, oilseeds, orchard crops, legumes and vegetables benefit from apiculture, increasing crop yield to the extent of 20%-80%.
- The report also says, India with the second largest arable land area, can increase substantially its existing output.



## LOOK AT THE STATISTICS ABOUT HONEY PRODUCTION IN INDIA!

- The honey production increased, as per the Ministry of Agriculture, from 51,000 tonnes in 2006-07 to 1,15,000 tonnes in 2018-19.
- India ranks 8<sup>th</sup> in the World in production. It is the 5<sup>th</sup> largest exporter.
- India exports about half of its output, fetching about Rs. 7,000 crore annually.





## APICULTURE CAN GIVE HIGH RETURNS WITH LESS INVESTMENT!

- Apiculture has the potential to support incomes with low initial investment.
- Each bee box costs Rs. 5,000. With 10 bee boxes per beneficiary, annual income of Rs. 30,000 can be achieved.
- Hence, apiculture is unique in providing livelihoods.

## NEED OF THE HOUR!

India must raise its bar on testing, apply the best techniques and integrate beekeeping with organic farming initiatives, which will bolster the economic viability of the latter!

## MAINTAINING STANDARDS OF HONEY IS THE REAL CHALLENGE!

- Honey is contaminated not only by the use of sugar syrup in processing, but also through pesticide and antibiotics use.
- The use of antibiotics such as terramycin and oxytetracycline to deal with bee-related diseases has raised questions in European markets, where residue standards are stringent.





# PLANTATION PROFILE IN INDIA ... 2019-20

|                                 | Tea       | Rubber    | Coffee   | Cardamom | Pepper    | Total     |
|---------------------------------|-----------|-----------|----------|----------|-----------|-----------|
| No. of growers/estates          | 1,59,190  | 13,10,537 | 3,63,366 | 30,000   | -         | 18,63,093 |
| Area (hectare)                  | 5,80,181  | 8,10,800  | 4,54,722 | 70,080   | Intercrop | 19,15,783 |
| Production (tonnes)             | 13,50,040 | 6,42,000  | 3,19,500 | 12,950   | 62,144    | -         |
| Yield (kgs/hectare)             | 2,284     | 1,489     | 765      | 427      | -         | -         |
| Value of crop (₹ crore)         | 21,196    | 8,121     | 10,736   | 2,122    | 2,534     | 44,709    |
| Export - quantity (tonnes)      | 2,56,100  | 4,551     | 3,53,853 | 2,250    | 16,840    | -         |
| (Percentage in production)      | 19.0      | 0.7       | 110.8    | 17.4     | 27.1      | -         |
| Export - value (₹ crore)        | 5,335     | 57.06     | 5,905    | 242.5    | 821       | 12,361    |
| Consumed within India (tonnes)  | 10,93,940 | 12,11,940 | -        | 10,700   | 45,304    | -         |
| Labour employed (lakhs)         | 12.60     | 4.93      | 6.60     | 0.30     | -         | 24.43     |
| Normal gestation period (years) | 3-4       | 6-7       | 5-7      | 2-3      | 3-4       | -         |
| Normal economic life (years)    | 60-80     | 25-35     | 40-60    | 12-15    | 30-35     | -         |





## COFFEE BOARD

- Coffee Board is under the Ministry of Commerce and Industry and is headquartered in Bengaluru.
- The main activity of the Board is enhancement of production, productivity, export promotion etc.
- Coffee marketing became a private sector activity in India since 1996 due to the economic liberalization.



**काँफी बोर्ड**  
**COFFEE BOARD**  
Government of India  
Ministry of Commerce & Industry

## TEA BOARD

- Tea Board was setup on 1<sup>st</sup> April, 1954.
- It is also under the Ministry of Commerce and Industry and is headquartered in Kolkata.



**टी बोर्ड भारत**

भारत सरकार वाणिज्य एवं उद्योग मंत्रालय के अधीन

**TEA BOARD INDIA**

Under Ministry of Commerce & Industry,  
Government of India



## SPICES BOARD

- Spices Board is also under the Ministry of Commerce and Industry and is headquartered in Kochi.
- It is responsible for the export promotion of 52 scheduled spices and development of cardamom.



**स्पाइसेस बोर्ड भारत**

वाणिज्य एवं उद्योग मंत्रालय, भारत सरकार

**SPICES BOARD INDIA**

Ministry of Commerce & Industry, Govt. of India







## A BRIEF ON TEA SECTOR IN INDIA

- India is the second largest producer of tea. The largest producer is China.
- Almost 70-80% is consumed in India i.e., only around 20-30% of tea produced is for export purposes.
- The per capita annual tea consumption is around 786 gram.
- Structural shift happened in recent decades in the form of emergence of small tea growers as a dominant force.
- In 2019, the tea estates segment share in total crop is just around 50%. To conclude, small tea growers share is increasing and the share of tea estates segment is reducing.



## Shrinking pie of arabica

■ Arabica (in tonnes) ■ Robusta (in tonnes) ● as % of total output



Source: Coffee Board



## MORE THAN 70% OF COFFEE PRODUCED IS FOR EXPORTS!

- More than 70% of India's coffee is for export purposes.
- 40% of the exported coffee is of six coffee varieties, that have a Geographical Indication tag.
- However, the share of Arabica decreased from around 35% to around 30% during the past two decades.





# IMPACT OF CLIMATE CHANGE ON PLANTATION CROPS



## 1 ON RUBBER PLANTATIONS



*There are clear indications that the extent of best suited areas for rubber cultivation will shrink in Kerala and the Konkan regions, whereas more areas may become better suited for rubber farming in the North-East in future.*

- Natural rubber growing regions have witnessed change in climate, particularly rising temperatures and altered rainfall patterns.
- Rising temperatures have varied impact on growth and yield in different agro- climatic regions.
- As temperature rises by 1 degree, the yield will go down by 10% or more in the traditional rubber growing regions of Kerala, but there is little adverse effect on rubber yield in North-East India.





# IMPACT OF CLIMATE CHANGE ON PLANTATION CROPS



## 2 ON COFFEE PLANTATIONS

**1** Karnataka is the largest producer of coffee in the country accounting for around two-thirds of the output, followed by Kerala.

**2** Heavy rains in recent years triggered flooding and landslides in the key coffee growing regions of Kodagu, Chikmagalur and Wayanad.

**3** In times of excess rains, coffee plantations are affected due to berry droppings.

**4** The ripened fruits on coffee bushes split and fall down to the earth, resulting in a crop loss for the growers.







# IMPACT OF CLIMATE CHANGE ON PLANTATION CROPS



## 2 ON COFFEE PLANTATIONS



5 Arabica plantations across the region also saw an increase of incidence of White Stem Borer, which is unique to India and research till now could not yield any tangible results.

6 Moreover, Arabica production has been witnessing decline over the past few decades.

*Since most of the coffee is grown under shade in India, it is insulated against climate change effects up to a certain degree.*





# IMPACT OF CLIMATE CHANGE ON PLANTATION CROPS



## 3 ON TEA PLANTATIONS



***Tea sector employs around 12-13 lakh people, followed by Coffee, with around 6-7 lakh and Rubber around 5 lakh.***

- 1 Severe frost attacks hit the production of tea leaves during winter.
- 2 The flare-up in temperatures during April-May affect the production of leaves.
- 3 Incidents of heavy rainfall within the short period sometimes result in huge erosion of top soil, with major implications for the hilly regions.





# IMPACT OF CLIMATE CHANGE ON PLANTATION CROPS



## 3 ON TEA PLANTATIONS



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- The growers towards the eastern slopes of Nilgiris are facing problems due to the decline in the number of rainy days and increase in the number of sunny days due to which flare-up of the pests is taking place.
- Red spider mite was a minor pest till recently. Now, it is spreading in the Nilgiris under the influence of prolonged dry weather.
- Moreover, tea made from leaves affected by red spider mite is very poor in quality.



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